

LEARNING ROBUST ACTIONS FROM REWARD MODELING

by

JACOB KRUSE

(Under the Direction of Prashant Doshi)

ABSTRACT

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INDEX WORDS: Robust Reinforcement Learning, Model-Based Reinforcement Learning, Offline Reinforcement Learning, Robotic Grasping, Regularization

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DEDICATION

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CHAPTER I

INTRODUCTION

Reinforcement Learning (RL) has achieved remarkable success in simulation, where agents can execute many actions and easily evaluate rewards. In simulated environments, large amounts of data can be generated quickly and safely, allowing policies to improve through extensive trial and error. For instance, (Vecerik et al., 2017) trains a robotic manipulator in simulation to perform a variety of manipulation tasks, while (Mnih et al., 2015) applies RL to teach an agent to play Atari games. These methods demonstrate the ability of RL algorithms to learn complex behaviors when there are large amounts of interaction data available to the agent.

Recently, RL has been increasingly applied to real-world tasks, motivated by the desire to tackle challenging problems beyond simulation. For example, (Smith et al., 2022) demonstrates how a legged robot can learn to traverse real world terrains, and (Wu et al., 2023) shows the application of their method to multiple physical platforms, including a quadruped robot, a robotic manipulator, and Sphero robot. Another prominent application of real-world RL is robotic grasping (Han et al., 2023), due to its widespread relevance in manipulation tasks such as warehouse automation and manufacturing. In these domains, robust grasping strategies are essential for practical deployment.

Despite this progress, the application of RL in the real world remains significantly more difficult than in simulation. The use of physical systems introduces uncertainty in sensing and actuation, while environmental factors introduce variability that further complicates learning. Additionally, robotic tasks

often involve high-dimensional continuous state and action spaces, leading to challenges associated with the curse of dimensionality. Data collection is also costly and time-consuming, as it requires executing actions on physical hardware. Furthermore, safety must be considered to avoid damaging equipment or causing injury. These challenges make it difficult to apply traditional RL methods, which typically rely on large amounts of accurate data obtained through online interaction.

Offline Reinforcement Learning is a popular strategy for addressing these limitations in physical systems. Offline RL methods learn policies entirely from pre-collected datasets without requiring additional environmental interaction during training. This greatly improves sample efficiency by enabling the algorithm to focus entirely on policy optimization rather than exploration. However, Offline RL introduces its own challenges, most notably distribution shift. Since the learned policy may select actions that differ from those present in the dataset, the value estimates for these actions cannot be evaluated reliably.

To mitigate this issue, many offline RL approaches incorporate techniques such as regularization and policy constraints. Regularization involves adding a term to the learning objective that encourages desirable behaviors in the learned policy, such as safer actions, improved exploration, or policy outputs that remain close to the collected data distribution. Incentivizing these behaviors can help stabilize training and reduce the risk of exploiting inaccuracies in value estimates.

Similarly, Maximum Entropy Reinforcement Learning (MaxEnt RL) (Haarnoja et al., 2017) alters the standard RL objective by incorporating an entropy term to the reward. This encourages the policy to maximize both expected return and entropy, promoting stochasticity in the policy and encouraging broader action distributions. While this resembles regularization with the additional term, MaxEnt RL fundamentally redefines the optimization objective. A well-known example is Soft Actor-Critic (SAC) (Haarnoja et al., 2018), which optimizes the MaxEnt objective and has demonstrated strong performance in continuous control tasks.

Inspired by these approaches, this work proposes an alternative objective that promotes robustness in the selected actions. A policy that chooses actions that perform well in ideal conditions but fail in practice due to noise from imperfect actuation, sensing, or environmental disturbances is undesirable. Therefore,

we encourage the policy to select actions that not only achieve high reward but also remain effective under small perturbations.

I.1 Contributions

In this paper, we introduce a framework that learns an approximate reward model and leverages it for iterative policy optimization with a novel robustness-oriented objective. Our approach is inspired by ideas from Soft Actor-Critic and Maximum Entropy Reinforcement Learning. The data collection strategy helps provide a diverse dataset for learning an accurate initial reward model. By learning an approximate reward function, we are able to efficiently and accurately predict rewards for a large number of state-action pairs. Focusing on action robustness during optimization encourages policies that are more tolerant to perturbations in real-world execution. Finally iterative deployment allows the method to obtain on-policy data and evaluate the effectiveness of the converged policy before further improvement. The main contributions of this work are:

- A new learning objective, the **I Don't Know Yet (IDKY)**, which encourages selection of actions that are both high-reward and robust to perturbations.
- Experiments demonstrating the method on a robotic sorting task, showing improved performance and robustness compared to standard policy optimization.
- An iterative learning algorithm that refines both the reward model and policy by combining expert, random, and on-policy data to efficiently improve grasping performance.

I.2 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2 - Related Work:** This chapter reviews prior work on Learning Grasp Success Models, Robust Reinforcement Learning, Offline Reinforcement Learning, and Model-Based Reinforce-

ment Learning. While these areas share similarities with the proposed method, they differ from the approach presented in this work.

- **Chapter 3 - Preliminaries:** This chapter introduces the foundational concepts necessary to understand the proposed method. Section 3.1 defines the *blank* framework used to model the grasping problem, while Section 3.2 describes how reward models can be learned using supervised learning techniques.
- **Chapter 4 - Methodology:** This chapter presents the proposed method in detail. Section 4.1 describes strategies used to learn an accurate reward model, including an effective data collection strategy and the use of ensemble networks capture uncertainty. Section 4.2 introduces the main algorithm for policy learning, which utilizes the newly defined IDKY objective and an iterative deployment strategy to progressively improve the the policy.
- **Chapter 5 - Experiments:** This chapter describes the experimental evaluation of the method. It includes details of the robotic grasping setup, datasets used, evaluation metrics, and results demonstrating the robustness of the proposed approach.
- **Chapter 6 - Conclusion:** This final chapter summarizes the contributions of the thesis, discusses the implications of the results, and outlines possible directions for future work.

CHAPTER 2

RELATED WORK

This chapter reviews the key areas of prior work that inspired the proposed approach. Section 2.1 discusses learning-based methods for robotic grasping, with a focus on models that predict grasp success from sensory input. Next, Section 2.2 introduces Robust Reinforcement Learning, which addresses challenges arising from uncertainty and adversarial effects. Section 2.3 describes Offline Reinforcement Learning, emphasizing learning from static datasets. Finally, Section 2.4 explains Model-Based Reinforcement Learning, highlighting techniques that leverage learned models. These areas provided the conceptual foundation and ideas that motivated the development of this work.

2.1 Learning Success Models for Grasping

Robotic grasping is a major area of study within robotics due to its wide use in manipulation tasks. As these tasks become more complex, traditional hand-engineered approaches are becoming increasingly ineffective. As a result, a substantial portion of research in robotic grasping has shifted toward learning-based methods that leverage datasets to model grasp outcomes. Many of these approaches focus on learning a grasp success model that predicts whether a proposed grasp action will succeed in the current state of the environment.

These models are trained to predict the success probability of candidate grasps using information such as RGB images, depth maps, object positions, or affordance representations. For example, after collecting data from 50,000 grasp attempts, (Pinto & Gupta, 2016) trained a Convolutional Neural Network that takes image patches as input and outputs the probability of grip success for a set of candidate grasp angles. Similarly, (Levine et al., 2018) learned a grasp prediction network that predicts the likelihood of a successful grasp from visual input and integrated this estimate within a continuous visual servoing controller.

Other work has explored the use of synthetic datasets to train these grasp evaluation models. After generating a large dataset of synthetic point clouds, (Mahler et al., 2017) learned a grip robustness function that evaluates candidate grasps and enables planning by selecting the highest-scoring action. This approach demonstrates how simulated data can be used to train these models while allowing them to generalize to real world tasks. Other methods, such as (Asif et al., 2019) and (Detry et al., 2011), learn these grasp success models from interaction data and utilize them for simple evaluation or to find high success actions.

The method in this paper follows this line of work by learning a grasp success model that maps states and actions to success probability. However, rather than using this learned model solely as an evaluation metric for grasp selection, the success predictor is interpreted as the expected reward function in this work. This reward model is then incorporated into the reinforcement learning framework to optimize an effective grasping policy.

2.2 Robust Reinforcement Learning

Robust Reinforcement Learning (Moos et al., 2022) encompasses a class of methods that extend the standard reinforcement learning objective to account for hostile environmental effects or adversarial factors. These approaches are especially useful for real world applications, where uncertainty and disturbances are inherent to the environment. The goal therefore is to learn a policy that achieves high expected reward while remaining effective in the presence of noise, adversaries, and uncertainty.

Prior work in Robust RL can be categorized based on the source of uncertainty that the method aims to address. The most common approaches focus on robustness with respect to uncertainty in the

transition dynamics or reward function. For example, (Bagnell et al., 2001) accounts for the stochastic nature of the environment by reformulating the reinforcement learning objective as a minimax problem, where the policy is optimized against the worst-case dynamics drawn from the set of possible transition probabilities. Additionally, (Xu & Mannor, 2006) learns a policy that maximizes a combination of a performance and robustness measure, where the robustness component captures the worst-case reward for the evaluated action. By optimizing against the worst-case outcomes in the reward or transition dynamics, these approaches produce policies that are robust but often conservative, sometimes resulting in reduced performance.

Many formulations that address transition and reward uncertainty are limited to tabular settings and do not easily extend to nonlinear function approximators or continuous state-action spaces. Another line of research in Robust RL instead focuses on robustness to adversarial or noisy actions, which can be applied in these more complex settings. The work of (Tessler et al., 2019) introduces the Noisy Action Robust MDP, where the standard policy is replaced with a joint policy π_{mix} that perturbs the intended action through an adversarial policy. The resulting optimization is formulated in a minimax framework for the mixed policy. Similarly, (Tan et al., 2020) defines a robust optimization objective in which adversarial perturbations are applied to actions in order to minimize the expected reward before policy optimization occurs. Much like the transition and reward methods, these methods aim to learn policies that perform well under worst-case action perturbations.

The objective proposed in this work draws a significant amount of inspiration from Robust RL techniques. However, rather than optimizing for the worst-case scenario like the methods described above, the approach is to optimize with an added term that quantifies the robustness of actions to perturbations. This formulation aims to produce policies that remain effective under environmental uncertainty without becoming overly conservative, thereby balancing robustness and performance.

2.3 Offline Reinforcement Learning

Offline Reinforcement Learning (Prudencio et al., 2023) focuses on learning a policy π entirely from a fixed dataset $\mathcal{D}$ that has been previously collected by an exploration or behavior policy π_β , without any additional interaction with the environment (see Figure 2.1). This approach contrasts with Online Reinforcement Learning, where the agent continually collects new experience while learning. Despite this distinction, the main objective of Offline RL is the same, which is to learn a policy that maximizes expected cumulative reward. Because training is performed on static, batched data, Offline RL is often more sample-efficient and can converge faster than Online RL, where interactions with the environment may be costly and slow down the learning process.

However, a central challenge with Offline RL is distribution shift. During training, the learned policy π may select actions or reach states that were rarely or never encountered in the dataset $\mathcal{D}$. This mismatch between the training data distribution and the states and actions encountered by the learned policy can result in unreliable value estimation, which in turn may lead to poor policy performance when deployed. To address this issue, prior work has developed a range of strategies. Techniques such as importance sampling (Precup et al., 2000) help correct the bias induced by off-policy evaluation, while regularization methods (Kumar et al., 2020) can encourage desirable behaviors in the learned policy that may not be sufficiently represented in the dataset $\mathcal{D}$. Similarly, policy constraints (Fujimoto et al., 2019) can be used to keep the learned policy π close to the dataset $\mathcal{D}$, improving the reliability of value estimation.

Although the method presented in this paper incorporates principles from Offline RL, it does not implement a pure offline approach. The learned policy π is subsequently deployed in the environment to acquire more data, which is then used to iteratively refine the reward model and policy. Despite this, the strategy of regularization formed a primary inspiration for this work. The combination of offline training with online data collection gives this method the efficiency of Offline RL while also addressing the limitations of static datasets.

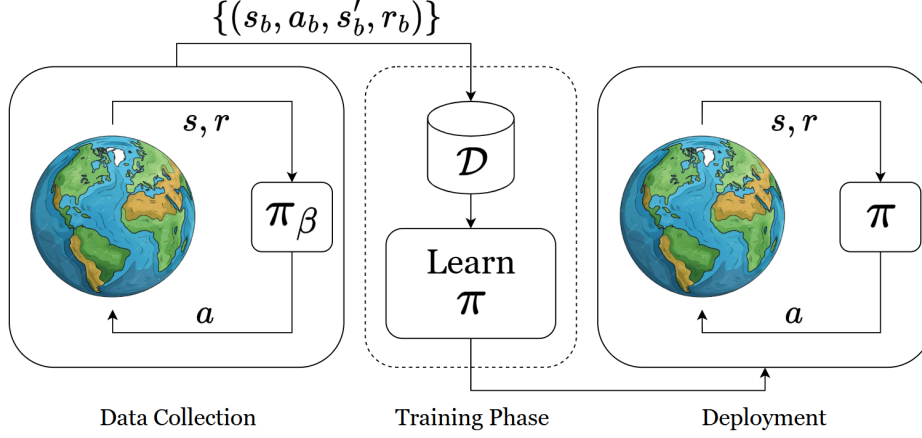


Figure 2.1: Offline Reinforcement Learning. An exploration policy π_β interacts with the environment to collect a dataset of state-action transitions and rewards, which is stored in the buffer $\mathcal{D}$. This dataset is used to train a policy π , which is then deployed in the environment without further interaction.

2.4 Model-Based Reinforcement Learning

Model-Based Reinforcement Learning (Luo et al., 2024) is a class of methods that learn a model of the environment’s dynamics or transition function, which predicts the resulting state of actions taken by the agent. The learned transition model allows the agent to simulate future states and generate synthetic data without interacting with the real environment. This is particularly useful in domains where real world interactions are costly, dangerous, or time consuming. Leveraging a learned model, these methods significantly improve sample efficiency since the agent can perform planning and optimization on a combination of real and simulated experiences. A basic Model-Based RL pipeline is illustrated in Figure 2.2.

A key challenge in Model-Based RL is model accuracy and uncertainty. Several approaches address this issue through uncertainty-aware modeling, such as ensemble methods which train multiple independent models. One approach, proposed in (Matsushima et al., 2020), trains an ensemble of K dynamics models and randomly selects one for each rollout, which helps reduce overfitting and increases robustness to individual errors. Additionally, (Yu et al., 2020) learns N reward models and estimates the reward as the

ensemble mean penalized by the maximum standard deviation to incentivize the policy to avoid regions of high uncertainty.

In addition to ensemble methods, many Model-Based RL methods adopt iterative model refinement and Dyna-style (Sutton, 1991) techniques to continuously improve the dynamics model and policy. Works such as (Feinberg et al., 2018) and (Nagabandi et al., 2018) follow this paradigm by alternating between model learning and data collection with the optimized policy to progressively improve performance. This leads to more robust and effective agents, especially in complex environments.

Although the proposed approach is related to Model-Based RL, it does not explicitly model a transition function. Instead, the method focuses on learning a reward model, which estimates the expected return of actions directly. Using the reward predictions, the policy can be optimized without the need for modeling state transitions. Despite this distinction, the approach adopts ideas from Model-Based RL, including the use of ensemble networks to capture uncertainty and iterative model improvement to enhance model accuracy.

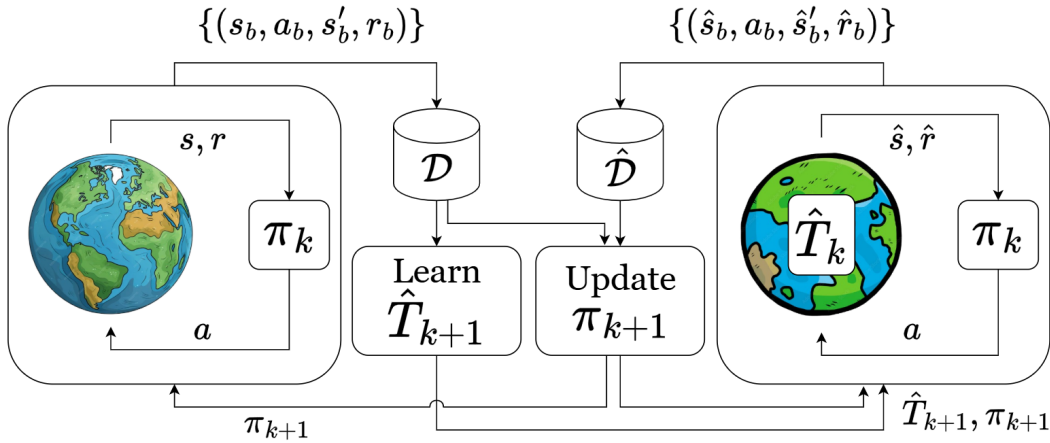


Figure 2.2: Model-Based Reinforcement Learning. A policy π_k is executed in the real world to collect data, which populates the real world buffer $\mathcal{D}$. This data is used to learn an approximate transition function $\hat{T}_k$. The learned transition function is used to generate a simulated dataset $\hat{\mathcal{D}}$, which is combined with the real-world dataset $\mathcal{D}$ to update the policy π_{k+1} . This iterative process improves sample efficiency by utilizing simulated data while grounding learning in real world experience.

CHAPTER 3

FOUNDATIONAL CONCEPTS

This chapter reviews the fundamental principles and methods that define this work, providing the necessary background for understanding the techniques presented later. Section 3.1 introduces Contextual Bandits, a framework for context-aware, action-selecting agents. Section 3.2 discusses Policy Gradient Reinforcement Learning, highlighting the basics of policy optimization with the gradient of the learning objective. Section 3.3 describes the process of modeling the expected reward function, emphasizing the estimation of unknown reward distributions through self-supervised learning. Finally, Section 3.4 introduces regularization in RL, explaining how it can improve the policies learned from static datasets. Together, these preliminaries establish the theoretical and practical foundation for the approach developed in this work.

3.1 Contextual Bandits

The Multi-Armed Bandit (Slivkins, 2024) framework is one of the most widely used models for decision making under uncertainty. In this setting, a single agent chooses from K arms or actions in each round and receives a reward from an unknown distribution associated with the chosen arm. After observing the reward, the agent repeats this process by selecting from the same set of arms, making it a sequential

problem with inherent uncertainty. The objective is to both learn the reward distribution of each arm and to maximize the expected reward.

A common extension of the Multi-Armed Bandit model is the Contextual Bandit framework (Langford & Zhang, 2007). Similar to the previous formulation, the agent chooses from a set of K arms or actions; however, in this case, the reward distribution now depends on an observed context. Despite this added structure, the goal of the Contextual Bandit problem remains the same as the Multi-Armed Bandit, to maximize reward in a stochastic and unknown environment. The Contextual Bandit (CB) can be formally defined with the tuple

$$\text{CB} = \langle \mathcal{X}, \mathcal{A}, R \rangle, \quad (3.1)$$

where $\mathcal{X}$ denotes the context space, $\mathcal{A}$ represents the action space, and R is the reward function. These components are explained below:

- **Context Space:** The context space $\mathcal{X}$ is the set of all possible contexts in the environment. In each round, a context $x \in \mathcal{X}$ is observed, representing the current state of the environment. The context x is drawn from an underlying context distribution ρ_x . The context may be discrete, continuous, or multi-dimensional, and is assumed to be fully observable.
- **Action Space:** The action space $\mathcal{A}$ is the set of all possible actions available to the agent. Given a context x , the agent selects an action $a \in \mathcal{A}$ and executes it in the environment. The action may also be discrete, continuous, or multi-dimensional.
- **Reward Function:** The reward function R is used to determine the reward r after each round. The reward function R is a function of the context x and action a and returns a distribution of real-valued rewards $R : \mathcal{X} \times \mathcal{A} \rightarrow \Delta_{\mathbb{R}}$. The observed reward r is then sampled from the reward function $r \sim R(\cdot|x, a)$ during each round. The underlying reward distribution induces an expected reward function $\mathcal{R}(x, a) = \mathbb{E}_{r \sim R(\cdot|x, a)}[r]$, which represents the average reward for each context-action pair.

In order to determine which actions to take in each context, the agent implements a policy π , which defines a mapping from the context space to the action space. A policy can either be stochastic or deterministic, depending on the desired behavior. A stochastic policy maps a given context x to a probability distribution over the actions $\pi : \mathcal{X} \rightarrow \Delta_{\mathcal{A}}$. To select an action, the agent samples the action a from the resulting distribution $a \sim \pi(\cdot|x)$. In the deterministic case, the policy maps each context to a single action $\pi : \mathcal{X} \rightarrow \mathcal{A}$, such that $a = \pi(x)$ with a probability of 1.

3.2 Policy Gradient Reinforcement Learning

Reinforcement Learning (RL) (Sutton, Barto, et al., 1998) is one of the most extensively studied paradigms in artificial intelligence. It can be described as the process of an agent learning rational, favorable behavior through interactions with its environment. A wide variety of techniques fall under the umbrella of RL, but they are generally categorized into two broad classes: Model-Based and Model-Free methods. In Model-Based RL (see Section 2.4), the agent tries to learn a model of the environment and uses this model for planning or predicting outcomes. In contrast, Model-Free RL bypasses the need for a model and aims to learn through direct interaction with the environment. Model-Free methods are highly popular due to their simplicity, robustness, and ability to scale to complex domains.

To take this a step further, Model-Free RL methods can be divided even further into Policy Gradient, Value-Based, and Actor-Critic techniques. In Policy Gradient approaches, the agent directly optimizes its policy by interacting with the environment and using the observed rewards to guide policy updates. These approaches are well-suited for continuous action spaces, as they focus solely on the policy, but are generally restricted to being on-policy due to the nature of policy optimization. In contrast, Value-Based RL methods focus on learning a function that estimates the value of states or actions, which is then used to guide action selection. These methods are typically more sample-efficient than Policy Gradient techniques because they can reuse past data, but they can struggle with large or continuous state and action spaces. Finally, Actor-Critic methods combine the strengths of Policy Gradient and Value-Based approaches. In this framework, the actor represents the policy and the critic refers to the value function. The feedback of

the critic is used to update the actor, leading to more stable and efficient learning compared to the either method alone.

Despite the wide variety of RL algorithms, the underlying learning objective remains the same: to find a policy π that maximizes the expected reward. The most general form of the learning objective is defined with the Markov Decision Process (MDP) framework (Bellman, 1957), and is commonly expressed as

$$J(\pi) = \mathbb{E}_{s_0 \sim \rho_s, a_t \sim \pi(\cdot|s_t), s_{t+1} \sim T(\cdot|s_t, a_t)} \left[\sum_{t=1}^{\infty} R(s_t, a_t) \right], \quad (3.2)$$

where the initial state s_0 is sampled from the state distribution ρ_s , actions a_t are sampled from the stochastic policy π , resulting states s_{t+1} are sampled from the stochastic transition function T , and rewards are given by the reward function R which is assumed to be deterministic.

To adapt this learning objective to the Contextual Bandit setting, some changes need to be made. Since Contextual Bandits involve no state/context transitions and no notion of sequential decision-making, the transition function T and the summation over all time steps in an episode can be removed. Additionally, the reward function is typically modeled as stochastic rather than deterministic, with observed rewards sampled as $r \sim R(\cdot|x, a)$. This can be handled either by taking an expectation over sampled rewards or, equivalently, by directly using the expected reward function $\mathcal{R}$, which simplifies the formulation while remaining consistent with the original objective.

On top of all this, let's consider the implementation of a deterministic policy, where each context x is mapped to a fixed action $a = \pi(x)$. Under these modifications, the learning objective becomes

$$J(\pi) = \mathbb{E}_{x \sim \rho_x} \left[\mathcal{R}(x, \pi(x)) \right], \quad (3.3)$$

where contexts x are sampled from the context distribution ρ_x , rewards are given by the expected reward function $\mathcal{R}$, and actions are deterministically selected by the policy π . Although Contextual Bandits may employ either stochastic or deterministic policies, this work focuses on deterministic policies; therefore, Equation 3.3 is adopted as the learning objective. Under this formulation, the optimal policy π^* is defined

as the policy that maximizes expected reward:

$$\pi^* = \arg \max_{\pi} J(\pi). \quad (3.4)$$

With the advancement of Deep Reinforcement Learning (Y. Li, 2017), it has become increasingly common to represent policies using neural networks. A neural network (Gurney, 2018) can be described as a parameterized function composed of multiple layers and interconnected nodes, where each layer applies weights, biases, and activation functions. After receiving an input, the network propagates it through its layers to produce an output. In the RL setting, the neural network acts as the policy, taking a state or context representation as input and producing an action or distribution over actions as output. The use of these networks enables highly expressive, nonlinear function approximation, allowing policies and value functions to handle high-dimensional and complex domains. When a policy is parameterized by a neural network with parameters ϕ , it is denoted as π_{ϕ} . The learning objective can then be written as a function of these parameters:

$$J(\phi) = \mathbb{E}_{x \sim \rho_x} [\mathcal{R}(x, \pi_{\phi}(x))]. \quad (3.5)$$

Importantly, the objective itself has not changed, but instead, it is now expressed in terms of the parameters ϕ that represent the policy. This shift is crucial, as it transforms the reinforcement learning problem into an optimization problem over the parameter space.

This change gives rise to the central idea behind Policy Gradient methods, which is to optimize the parameters ϕ using the gradient of the objective. More specifically, the goal is to iteratively update the parameters ϕ in the direction of the gradient $\nabla_{\phi} J(\phi)$, which quantifies how changes in the parameters affect the expected reward. For the Contextual Bandit setting, the deterministic policy gradient can be expressed as a modification of the original formulation in (Silver et al., 2014):

$$\nabla_{\phi} J(\phi) = \mathbb{E}_{x \sim \rho_x} [\nabla_{\phi} \pi_{\phi}(x) \cdot \nabla_a \mathcal{R}(x, a)|_{a=\pi_{\phi}(x)}]. \quad (3.6)$$

This expression differs from the original deterministic policy gradient formulation because it is adapted to the Contextual Bandit setting rather than the MDP framework (Bellman, 1957). In a standard MDP, the policy gradient depends on the Q-function $Q(s, a)$, which represents the total discounted sum of future rewards starting from state s and taking action a . The Q-function captures transition dynamics and is required to propagate future rewards back to earlier actions. However, Contextual Bandits have no state or context transitions, so the policy does not need to account for future outcomes. As a result, the Q-function Q can be replaced by the immediate expected reward function $\mathcal{R}$, allowing the deterministic policy gradient to be applied directly to Contextual Bandits.

Policy Gradient methods use the gradient of the learning objective to directly optimize the policy parameters. More specifically, the algorithm performs gradient ascent, adjusting the policy parameters in the direction that increases the expected reward. At each update step, the parameters are adjusted according to

$$\phi_{k+1} = \phi_k + \lambda \cdot \nabla_{\phi_k} J(\phi_k), \quad (3.7)$$

where λ is the learning rate. This update effectively pushes the policy toward actions that yield higher rewards, reinforcing behaviors that are more likely to succeed.

In practice, the gradient of the learning objective cannot be calculated exactly because it depends on the full distribution of contexts and rewards, which is generally unknown. Instead, it must be estimated from sampled interactions with the environment. This can be done by executing the current policy π_ϕ in the real-world and adding the observed context, action, and reward tuples into a dataset $\mathcal{D}_{\pi_\phi} = \{(x_b, a_b, r_b)\}_{b \in \mathcal{D}_{\pi_\phi}}$. This dataset provides a finite sample of the policy's behavior and its effects on the environment, which can be used to estimate the gradient. The deterministic policy gradient can then be approximated as

$$\nabla_\phi J(\phi) \approx \frac{1}{\|\mathcal{D}_{\pi_\phi}\|} \sum_{b \in \mathcal{D}_{\pi_\phi}} \nabla_\phi \pi_\phi(x_b) \cdot \nabla_a \mathcal{R}(x_b, a_b)|_{a_b = \pi_\phi(x_b)}. \quad (3.8)$$

This expression is an unbiased Monte Carlo estimate of the true gradient. Although this estimate does not yield the exact gradient, it is sufficiently accurate to guide parameter updates in Policy Gradient methods, allowing the agent to iteratively improve its policy.

3.3 Reward Modeling

The reward function R for a Contextual Bandit, defined in Section 3.1, is stochastic and returns a distribution over real-valued rewards. After taking action a in context x , the observed reward r is sampled from this distribution $r \sim R(\cdot|x, a)$. The expected reward function $\mathcal{R}$ can be used equivalently to represent the sampled rewards, but it instead returns a scalar value. Because of this, if the expected reward function is known, the policy can use it directly for optimization by calculating the gradient with Equations 3.6 or 3.8.

In many practical settings, however, the true reward function is unknown and must be inferred through interactions with environment. One approach is to learn a policy directly from past experience and utilize importance sampling (Precup et al., 2000). Since this experience is collected under an exploration policy π_β , whose action probabilities differ from the policy π_k being optimized, the expected reward cannot be accurately estimated without adjustment. Importance sampling addresses this by weighting observed actions according to how likely the trained policy π_k would select them, thereby producing a more accurate estimate of the expected reward under the target policy π_k .

Although importance sampling is powerful, it struggles with deterministic policies and continuous action spaces. As an alternative, one can use the collected data to learn a model that approximates the expected reward function $\hat{\mathcal{R}}$, and use this in place of the true expected reward function $\mathcal{R}$ for policy optimization. Under the exploration policy π_β , the agent observes tuples of the form (x_b, a_b, r_b) where x_b is the context, $a_b \sim \pi_\beta(\cdot|x_b)$ is the action taken, and r_b is the observed reward. These interactions can be used to generate a dataset $\mathcal{D} = \{(x_b, a_b, r_b)\}_{b \in \mathcal{D}}$. Modeling the approximate reward function then becomes a Supervised Learning problem.

In Supervised Learning (Cunningham et al., 2008), the goal is to obtain a model that predicts the true labels of all input-output pairs in a dataset. To measure the discrepancy between the true labels y corresponding to input x and predicted labels $\hat{y}$ from the approximate model $\hat{f}(x)$, a loss function $\mathcal{L}(y, \hat{f}(x))$ is introduced (C. Li et al., 2025). The empirical risk $\mathcal{L}(\mathcal{D}, \hat{f})$ or the average loss of the model over an entire dataset is then defined as

$$\mathcal{L}(\mathcal{D}, \hat{f}) = \frac{1}{\|\mathcal{D}\|} \sum_{b \in \mathcal{D}} \mathcal{L}(y_b, \hat{f}(x_b)). \quad (3.9)$$

This quantity measures the performance of the model on the observed data. The model that best fits the data $\hat{f}^*$ is obtained by minimizing the empirical risk:

$$\hat{f}^* = \arg \min_{\hat{f}} \mathcal{L}(\mathcal{D}, \hat{f}). \quad (3.10)$$

In practice, it is very common to represent the model using a neural network. For a model $\hat{f}_\theta$ parameterized by θ , the loss function can be rewritten as $\mathcal{L}(y, \theta)$, so that it is now a function of the true labels y and the parameters θ that represent the approximate model. With this change, Equation 3.10 is reframed as finding the parameters that minimize the empirical risk. This optimization problem is typically solved with gradient descent, where the parameters are iteratively updated in the direction of the negative gradient of the empirical risk. The parameters θ are updated according to the following rule:

$$\theta_{k+1} = \theta_k - \lambda \cdot \nabla_{\theta} \mathcal{L}(\mathcal{D}, \theta_k), \quad (3.11)$$

where λ is the learning rate.

In the setting of learning a model to approximate the expected reward function, the goal is to predict the true expected reward for each context-action pair. This means the reward model is trained by minimizing the loss between the true observed rewards and predicted expected rewards: $\mathcal{L}(r_b, \hat{\mathcal{R}}_\theta(x_b, a_b))$. Because rewards are obtained directly from interactions with the environment, no manual annotation

is required. In this case, reward modeling can be viewed as a form of Self-Supervised Learning (Jaiswal et al., 2020), an effective branch of Supervised Learning that avoids the cost of labeling large datasets. For a self-annotated dataset $\mathcal{D}$, the optimal parameters θ of the approximate expected reward function $\hat{\mathcal{R}}$ are obtained by iteratively minimizing the empirical risk over a sampled mini-batch B :

$$\theta^* = \arg \min_{\theta} \frac{1}{\|B\|} \sum_{b \in B} \mathcal{L}(r_b, \hat{\mathcal{R}}_{\theta}(x_b, a_b)). \quad (3.12)$$

Once trained, this model can be used for policy optimization, enabling sample-efficient learning without access to the true expected reward function.

When applying Supervised Learning, it is important to choose a loss function that is appropriate for the given problem. In the context of learning an approximate expected reward function $\hat{\mathcal{R}}$, the model is trying to capture the true underlying stochastic reward function R , which returns a probability distribution over discrete reward outcomes. Because of this, Cross-Entropy (CE) loss (Mao et al., 2023) is suitable, as it is widely used for probability models and classification tasks. The most general form of CE loss is Categorical Cross-Entropy loss $\mathcal{L}_{CCE}$, which is defined as

$$\mathcal{L}_{CCE}(\theta) = - \sum_{c \in C} y_c \cdot \log(\hat{y}_c), \quad (3.13)$$

where C is the set of categories for the labels. The equation above is for a single-instance loss, so the corresponding empirical risk $\mathcal{L}_{CCE}$ for a mini-batch B is

$$\mathcal{L}_{CCE}(\theta) = - \frac{1}{\|B\|} \sum_{b \in B} \sum_{c \in C} y_{b,c} \cdot \log(\hat{y}_{b,c}). \quad (3.14)$$

Minimizing this objective encourages the reward model to assign high probability to observed rewards and low probability to others, effectively learning a distribution over rewards.

In many practical settings, the reward is binary, taking one of two labels (success or failure). In this case, Binary Cross-Entropy (BCE) loss is appropriate. The per-sample BCE loss $\mathcal{L}_{BCE}$ is

$$\mathcal{L}_{BCE}(\theta) = -y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y}), \quad (3.15)$$

and the corresponding empirical risk $\mathcal{L}_{BCE}$ over a mini-batch B is

$$\mathcal{L}_{BCE}(\theta) = -\frac{1}{\|B\|} \sum_{b \in B} y_b \cdot \log(\hat{y}_b) + (1 - y_b) \cdot \log(1 - \hat{y}_b). \quad (3.16)$$

After training a reward model with BCE loss, the output represents the probability of receiving each of the two possible reward outcomes. For a binary reward system with values of 0 or 1, the expected reward simplifies to the probability of receiving 1:

$$\mathbb{E}_{r \sim R(\cdot|x,a)}[r] = 0 \cdot Pr(0|x, a) + 1 \cdot Pr(1|x, a) = Pr(1|x, a)$$

Since a reward of 1 is generally associated with a successful action, the reward model can be interpreted as outputting the probability of success.

3.4 Regularization in Offline Reinforcement Learning

Although Offline Reinforcement Learning is effective, it is not without limitations. A key challenge arises from the fact that optimization is performed on a static dataset. This can lead to overfitting, where the learned policy performs well on the observed data but fails to generalize to unseen situations when deployed in the real world. These failures arise because the policy exploits biases in the dataset, which may be incomplete or does not sufficiently cover the relevant state and action spaces. This is particularly problematic because Offline RL does not have the ability to correct these errors with environmental feedback like standard Online RL algorithms.

To address these limitations, a common approach is to incorporate regularization into the learning process. In Offline RL, regularization refers to the addition of an extra term or function to the standard RL objective. This term is designed to penalize or encourage specific states, actions, or characteristics, effectively shaping the learned policy towards behaviors that are not fully represented in the dataset. The main idea is that the dataset is limited or biased, and the regularization term serves as a corrective mechanism, guiding the policy towards more favorable outcomes. Formally, for a policy π_ϕ parameterized by ϕ and a regularization function $\mathcal{R}(\cdot)$, the regularized objective can then be written as

$$J(\phi) = \mathbb{E}_{x \sim p_x(\cdot)} [\mathcal{R}(x, \pi_\phi(x))] + \mathcal{R}(\cdot), \quad (3.17)$$

where the regularization term $\mathcal{R}(\cdot)$ can depend on any relevant variables, including the context x , action a , or even the policy π_ϕ itself.

The regularization function $\mathcal{R}(\cdot)$ is highly versatile and can be applied in multiple ways. For instance, adding a positive entropy term to the objective promotes stochasticity and encourages exploration in the policy, which is exemplified in (Haarnoja et al., 2018). Alternatively, regularization can be used to constrain the learned policy to remain close to the dataset distribution. An example from (Nachum et al., 2019) is the use of a negative f -divergence term, which penalizes deviations from the dataset and ensures the policy remains supported by the data. Overall, regularization is a powerful tool that can induce desirable behaviors and compensate for the limitations of Offline RL, all while maintaining computational and sample efficiency.

CHAPTER 4

METHODOLOGY

This chapter presents the proposed approach for learning a policy using an approximate expected reward function. Section 4.1 describes how the reward model is learned, emphasizing strategies to improve accuracy and reliability of the expected reward function approximation. Section 4.2 then introduces an enhanced policy optimization framework, detailing ways to improve performance of the learned policies. These components form an effective approach for leveraging limited reward signals efficiently while maintaining stability and performance in uncertain and costly environments.

4.1 Approximating and Accurate Reward Model

If the true expected reward function $\mathcal{R}$ for a Contextual Bandit is unknown, a learned approximation of the expected reward function $\hat{\mathcal{R}}$ can be used in its place to directly apply policy gradient methods. As detailed in Section 3.3, learning a reward model consists of utilizing a dataset $\mathcal{D}$ defined as $\mathcal{D} = \{(x_b, a_b, r_b)\}_{b \in \mathcal{D}}$. The reward serves as the output label, and self-supervised learning is performed on the context-action inputs to predict the corresponding expected reward.

Although this basic process can produce effective results, additional strategies are employed to further improve the accuracy and robustness of the approximate reward model. First, an effective data collection strategy is used to construct a dataset with broad coverage of the context-action space and a diverse range

of reward values to guide model learning. Second, the approximate expected reward function is modeled with an ensemble of networks, leading to improved predictive performance and introducing a mechanism to quantify and penalize uncertainty. Together, these components enable the learned expected reward function to produce more accurate and reliable reward estimates for context-action pairs. This accuracy is crucial, as the approximate expected reward model is subsequently used to guide and optimize the policy.

4.1.1 Initial Data Collection Strategy

To produce an accurate model with supervised learning, it is typically crucial to train with large and representative datasets, as it enables the model to capture the relationship between inputs and outputs across a wide range of scenarios. However, in many real world domains, producing large-scale datasets is often time-consuming and expensive, especially when manual annotation is required. Because of this, when working with limited data, it is essential to maximize the diversity of the training dataset (Gong et al., 2019). This diversity improves coverage of the input space and allows the model to generalize effectively without access to huge quantities of data.

Reinforcement learning is commonly applied to robotics despite the high cost of data collection, as it provides a natural framework for learning from interaction. In practice, many of these systems allow for human intervention through teleoperation or external control, which can be used to produce expert demonstrations that yield high reward behaviors. This gives rise to learning methods like Imitation Learning (Hussein et al., 2017), or more specifically Kinesthetic Teaching (Billard et al., 2006), where policies are trained to replicate the demonstrated actions. While these methods can produce effective policies, their goal is to mimic the demonstrated behavior rather than to infer and optimize an underlying reward function. As a result, they are limited to the expert data distribution and may fail to generalize or improve upon it.

Nevertheless, human-generated data provides a valuable source of information that can be leveraged beyond pure imitation. To construct a dataset that is sufficiently diverse, the proposed approach forms an initial dataset $\mathcal{D}_0$ by combining data generated by a human expert policy π_E and a random exploration

policy π_β . The term "expert" is used loosely here; while these actions should produce high rewards, they do not need to be strictly optimal. These samples provide a positive learning signal to identify desirable regions of the reward space for policy optimization. In contrast, the exploration policy π_β is likely to produce lower-reward actions, but its primary purpose is to ensure broad coverage of the context and action space. This exploratory data exposes the model to a wider range of outcomes, including failure cases, which improves overall data diversity. By combining expert and exploratory data, the resulting dataset achieves both reward diversity and context-action coverage. This balance is important for learning an approximate expected reward function that can be reliably used for downstream policy optimization.

4.1.2 Leveraging Ensemble Networks

When a dataset lacks sufficient diversity to adequately cover the input space, a learned model cannot be trusted in under-represented regions. While a narrowly distributed dataset will yield strong performance on in-distribution samples, this becomes problematic when the model is utilized for out-of-distribution samples. Specifically, if a model for the expected reward function is learned and used to guide policy learning, the policy may explore these poorly represented regions, resulting in unreliable reward estimates and degraded performance.

A common strategy in supervised learning to mitigate this issue is to train an ensemble of models rather than relying on a single predictor. One simple approach is to average predictions across the ensemble, which often yields smoother and more accurate estimates. More importantly, disagreement between the models can be used as a measure of uncertainty. Some methods utilize the model with worst-case prediction to incorporate this uncertainty, while others compute the variance across models to penalize estimates. Either way, ensemble methods can significantly improve robustness in the learned model.

In this work, an ensemble of N independently parameterized expected reward functions $\bar{\mathcal{R}}_{\theta_1}, \dots, \bar{\mathcal{R}}_{\theta_N}$ are trained. Each model is initialized with different parameters θ_i and trained on independently sampled mini-batches, which encourages diversity among the learned networks. The ensemble is used to compute

the mean of the expected reward μ_θ and the standard deviation σ_θ , defined as

$$\mu_\theta(x, a) = \frac{1}{N} \sum_{i=1}^N \bar{\mathcal{R}}_{\theta_i}(x, a), \quad (4.1)$$

$$\sigma_\theta(x, a) = \sqrt{\frac{1}{N} \sum_{i=1}^N (\bar{\mathcal{R}}_{\theta_i}(x, a) - \mu_\theta(x, a))^2}. \quad (4.2)$$

The mean μ_θ serves as the primary estimate of the expected reward, while σ_θ captures the level of disagreement across the ensemble. High variance indicates that the input lies in a region that is poorly represented in the dataset, which means the current prediction is less reliable.

To account for this uncertainty, the uncertainty-penalized reward $\hat{\mathcal{R}}_\theta$ is defined as

$$\hat{\mathcal{R}}_\theta(x, a) = \mu_\theta(x, a) - \beta \cdot \sigma_\theta(x, a), \quad (4.3)$$

where β is the uncertainty coefficient, a hyperparameter that controls the strength of the penalty. When used for policy optimization, larger values of β will lead to more conservative approximations of the expected reward, discouraging the policy from selecting actions in regions where the reward model is uncertain. This formulation can be interpreted as a lower confidence bound on the predicted expected reward. The training procedure for the ensemble is outlined in Algorithm 1.

Algorithm 1 Train Expected Reward Ensemble

Inputs: Dataset $\mathcal{D}$, Number of expected reward networks N , Uncertainty coefficient β

Output: Expected reward model $\hat{\mathcal{R}}_\theta$

- 1: Initialize N expected reward networks $\bar{\mathcal{R}}_{\theta_1}, \dots, \bar{\mathcal{R}}_{\theta_N}$ with random parameters $\theta_1, \dots, \theta_N$
 - 2: **for** $t = 1$ to T **do**
 - 3: **for** $n = 1$ to N **do**
 - 4: Sample an independent mini-batch: $B_n \subset \mathcal{D}$
 - 5: Update parameters with empirical risk: $\mathcal{L} = \frac{1}{\|B_n\|} \sum_{b \in B_n} \mathcal{L}(r_b, \bar{\mathcal{R}}_{\theta_n}(x_b, a_b))$
 - 6: **end for**
 - 7: **end for**
 - 8: Calculate uncertainty-penalized reward model: $\hat{\mathcal{R}}_\theta = \mu_\theta - \beta \cdot \sigma_\theta$
 - 9: **return** $\hat{\mathcal{R}}_\theta$
-

4.2 Improved Policy Optimization

After learning an approximate expected reward function, the next step is to perform policy optimization. This is typically achieved by iteratively updating a parameterized policy in the direction of the policy gradient (See Section 3.2). While standard policy gradient methods have demonstrated strong performance, they often rely on the assumption that the reward signal is accurate and that maximizing expected reward is sufficient to produce effective behavior.

Because the reward model is approximated, additional techniques are introduced to improve the reliability and robustness of the optimization process. First, a robustness-aware objective is proposed, which encourages the policy to select high reward and perturbation-resilient actions. Second, an iterative learning procedure is adopted, in which a policy trained offline is repeatedly deployed to collect new data, enabling continuous refinement of the reward model and policy. Collectively, these strategies lead to a more stable and reliable optimization process, resulting in improved policy performance. This is particularly important when relying on an approximate expected reward function trained from limited or imperfect data.

4.2.1 Perturbed Action Robustness Objective

In reinforcement learning, the main objective is to maximize expected reward, as formalized in Equation 3.3. In many scenarios, this objective is sufficient to produce intelligent agents. However, in real-world settings, optimizing expected reward alone may lead to suboptimal behavior. This can be attributed to several factors, including poor exploration, incomplete data, or uncertainty in the environment.

To address these limitations, various techniques can be applied. One common approach is regularization (See Section 3.4), where an additional term is incorporated into the RL objective to arouse certain behaviors in the policy being optimized. For example, regularization can promote improved exploration or bias the policy toward regions supported by reliable data. Regardless of the desired traits, the resulting agent is often more successful than one trained with the standard learning objective alone.

Another line of work focuses on explicitly handling uncertainty through Robust RL, which is explained in Section 2.2. These methods aim to account for uncertainty in different components of the RL problem, including the reward function and action execution. For instance, policies may be optimized with respect to the worst-case reward function or under adversarial perturbations to the actions. While such approaches produce robust agents, they can also lead to overly conservative policies due to their emphasis on the worst-case outcomes.

Both of these techniques were considered when addressing uncertainty in both the environment and the approximate expected reward function. To mitigate these issues, a new learning objective is introduced to promote robustness to action perturbations. First, a robustness measure, called the **Perturbed Action Robustness (PAR)** function, is defined as

$$\eta(x, a) \triangleq \frac{1}{Z} \cdot \int_{-\delta_l}^{\delta_l} \hat{\mathcal{R}}(x, a + \delta) d\delta, \quad (4.4)$$

where δ_l is the bounds of the action perturbations and Z is a normalizing constant. The normalizer is commonly defined as

$$Z \triangleq \int_{-\delta_l}^{\delta_l} r_{\max} d\delta, \quad (4.5)$$

where $r_{\max}$ is the maximum possible reward. By normalizing η with $\frac{1}{Z}$, the output remains within the bounds $[0, 1]$. Without normalization, the PAR function corresponds to the cumulative expected reward of an action as it is perturbed to the bounds, providing a measure of the action’s local robustness. Intuitively, it measures the effectiveness of nearby actions, assigning larger values to actions that lie in high reward regions. This robustness function is incorporated into the standard objective to obtain

$$J(\phi) = \mathbb{E}_{x \sim \rho_x(\cdot)} \left[\hat{\mathcal{R}}(x, \pi_\phi(x)) + \alpha \cdot \eta(x, \pi_\phi(x)) \right], \quad (4.6)$$

where α is the robustness coefficient, a parameter that determines the importance of the PAR function. This objective encourages the policy π_ϕ to select actions that not only achieve high expected reward, but those that are also stable under action perturbations. As a result, the learned policy π_ϕ is less likely to

exploit narrow peaks in the reward landscape and is more likely to converge to broad, high reward regions. The standard RL objective is recovered in the limit as $\alpha \rightarrow 0$.

The corresponding policy gradient is

$$\nabla_{\phi} J(\phi) = \mathbb{E}_{x \sim \rho_x(\cdot)} [\nabla_{\phi} \pi_{\phi}(x) \cdot \nabla_a (\hat{\mathcal{R}}(x, a) + \alpha \cdot \eta(x, a))|_{a=\pi_{\phi}(x)}]. \quad (4.7)$$

Much like the standard deterministic policy gradient shown in Equation 3.6, it is often intractable to calculate this expectation exactly. However, it can be estimated with Monte Carlo sampling, exemplified by the following

$$\nabla_{\phi} J(\phi) \approx \frac{1}{\|B\|} \sum_{b=1}^B \nabla_{\phi} \pi_{\phi}(x_b) \cdot \nabla_a (\hat{\mathcal{R}}(x_b, a_b) + \alpha \cdot \eta(x_b, a_b))|_{a_i=\pi_{\phi}(x_i)}, \quad (4.8)$$

where B represents the sampled batch of contexts. This approximation provides an effective learning signal to guide policy optimization under the proposed objective.

4.2.2 Iterative Deployment of Trained Policy

One of the major problems with Offline RL (Section 2.3) is its inability to improve its policy through interactions with the environment. Because learning is constrained to a fixed dataset, the learned policy is strongly dependent on the data collected by the exploration policy. When this data is incomplete or does not provide sufficient coverage of the context and action spaces, optimization may lead the policy to regions that appear optimal under the data distribution, but when deployed in the true environment, the policy underperforms.

Similarly, a major problem with Model-Based RL, which is detailed in Section 2.4, is the accuracy of the learned dynamics or reward model. Because these models are trained on previously collected experience, they inherit the same limitations as policies learned using Offline RL. In regions where data is sparse or absent, model predictions become unreliable. This is particularly problematic when policy updates shift the agent towards these out-of-distribution states, where the reward estimates are likely inaccurate.

To tackle these coupled issues, the proposed method combines offline training with iterative online deployment for policy evaluation and model refinement. After optimization under the modified objective described above, the resulting policy π_{ϕ_i} is executed in the environment to generate a new dataset $\mathcal{D}_{\pi_{\phi_i}}$. To improve exploration and to evaluate robustness, Gaussian noise is added to the continuous actions of the deterministic policy π_{ϕ_i} during deployment. The collected context-action-reward tuples are then aggregated with the existing dataset as follows:

$$\mathcal{D}_{i+1} = \mathcal{D}_i \cup \mathcal{D}_{\pi_{\phi_i}}, \quad (4.9)$$

resulting in a progressively expanding dataset that combines random exploration, expert data, and newly collected on-policy experience.

This aggregated dataset is subsequently used to retrain the expected reward model, allowing it to adapt to regions of the context-action space induced by the current policy. As a result, the expected reward approximator becomes progressively more accurate in high relevance regions. The improved expected reward model is then used re-optimize the policy, either reinforcing the current actions or steering the policy away from them. This iterative refinement helps mitigate errors arising from both the policy and the learned model, resulting in agent that is both sample and computational efficient while

remaining effective. The overall procedure is presented in Figure 4.1 and summarized in Algorithm 2.

Algorithm 2 Policy Optimization

Inputs: Number of reward networks N , Robustness coefficient α , Uncertainty coefficient β

Output: Policy π_ϕ

- 1: Initialize policy network π_ϕ with random parameters ϕ
 - 2: Fill initial dataset $\mathcal{D}_0$ with data from exploration policy π_β and from expert policy π_E
 - 3: **for** $i = 0$ to I **do**
 - 4: $\hat{\mathcal{R}}_\theta \leftarrow \text{TrainRewardModel}(\mathcal{D}_i, N, \beta)$
 - 5: **for** $t = 1$ to T **do**
 - 6: Sample mini-batch from current dataset: $B \subset \mathcal{D}_i$
 - 7: Update parameters with gradient:

$$\nabla_\phi J(\phi) = \frac{1}{\|B\|} \sum_{b \in B} \nabla_\phi \pi_\phi(x_b) \cdot \nabla_a (\hat{\mathcal{R}}_\theta(x_b, a_b) + \alpha \cdot \eta(x_b, a_b))|_{a_b = \pi_\phi(x_b)}$$
 - 8: **end for**
 - 9: Execute π_{ϕ_i} and add interactions to $\mathcal{D}_{\pi_{\phi_i}}$
 - 10: Update dataset: $\mathcal{D}_{i+1} = \mathcal{D}_i \cup \mathcal{D}_{\pi_{\phi_i}}$
 - 11: **end for**
 - 12: **return** π_ϕ
-

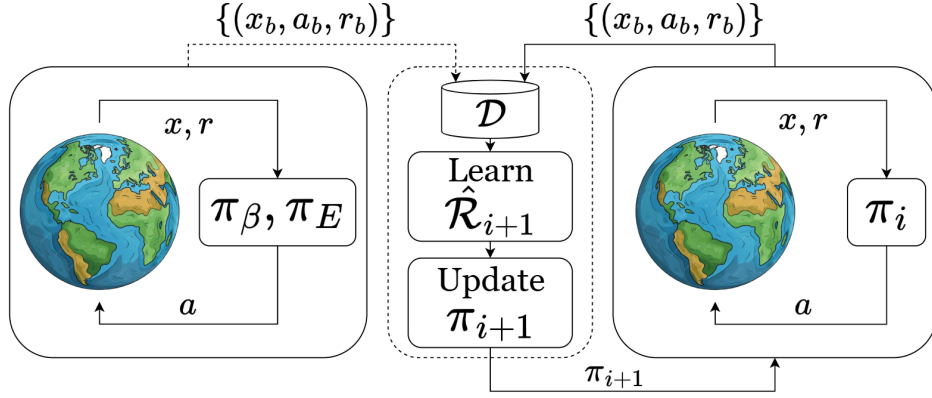


Figure 4.1: Iterative Reward Model and Policy Training. An exploration policy π_β and expert policy π_E interact with the environment to collect context-action-reward tuples, which are stored in the initial dataset $\mathcal{D}$. This dataset is used to train an approximate expected reward model $\hat{\mathcal{R}}_i$, which guides the optimization of a policy π_i . The learned policy π_i is deployed in the environment to generate on-policy data and adds it to the dataset $\mathcal{D}$. The reward model $\hat{\mathcal{R}}_{i+1}$ is updated with the augmented dataset, and a new policy π_{i+1} is optimized. This process repeats iteratively, progressively improving the reward model and policy.

CHAPTER 5

EXPERIMENTS

To evaluate the proposed method, experiments were done on a sorting task for recyclable items with a robotic manipulator. The following experiments are the main contributions of this paper.

5.1 Experimental Setup

This section describes the experimental framework used to evaluate the proposed method. The task setup is detailed, including the robotic and sensing hardware, as well as the overall behavior of the agent within the environment. Furthermore, the perception of the workspace state and its transformation into both the true and encoded context representations is explained. And finally, the action and reward definitions are described in detail. All of these components define a complete system in which the robot can apply the proposed approach to a real-world grasping task for sorting recyclable items.

5.1.1 Task Description

The proposed task is to autonomously sort recyclable items within a constrained workspace. The item set consists of aluminum cans and plastic bottles with varying degrees of deformation, which introduces variability in shape and grasp difficulty. The object manipulation is performed with a KUKA LBR iiwa 3000 collaborative robotic arm equipped with an Ubiros Gentle Pro gripper at the end effector.

The context of the workspace is captured using an overhead Oak-D S2 RGB-D camera, which provides synchronized color and depth images of the scene. These images are processed and encoded to form the input to the learned policy. The policy outputs continuous control actions for the manipulator, specifying the relative end effector placement. After executing the policy action with the corresponding motion trajectory, the robot automatically closes the gripper to secure the object, moves to the recycling bin, and opens the gripper to release any grasped object. An action is considered successful if the robot is able to securely grasp the assigned object and transport it to the recycling bin. The experimental setup is illustrated in Figure 5.1 and the overall sorting procedure is summarized in Algorithm 3.

Algorithm 3 Sorting

Input: Learned policy π_ϕ

```

1: while items in workspace do
2:   Camera captures RGB and Depth images of the workspace
3:   Images are processed to obtain context  $x$  and encoded context  $\hat{x}$ 
4:   Robot moves to "Home" position
5:   Determine action from policy:  $a \leftarrow \pi_\phi(\hat{x})$ 
6:   Robot executes action  $a$  by moving to relative end effector position
7:   Robot closes gripper to grab item
8:   Robot moves to "Bin" position
9:   Robot opens gripper to drop item
10: end while

```

5.1.2 Context Estimation

The context of the workspace is obtained using images from an overhead RGB-D camera. When the current context is requested, the camera captures an RGB image and a corresponding depth image of the workspace. The RGB image is passed through a YOLO v11 instance segmentation model, which produces pixel-wise masks of the aluminum cans and plastic bottles present in the scene. The YOLO model was trained on a combination of publicly available datasets containing annotated images of general cans and bottles, as well as a custom dataset containing images collected directly from the workspace with manual annotation. Incorporating general and task-specific data enables the model to generalize and maintain accuracy for specific objects with the instance segmentation procedure.

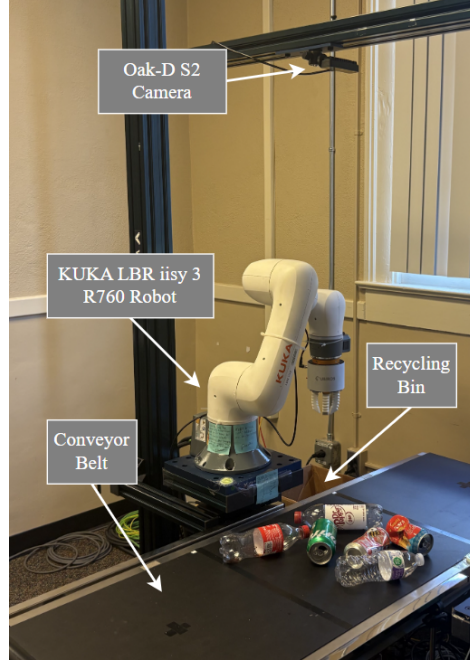


Figure 5.1: Experimental Setup. A KUKA robotic manipulator is used to grab aluminum cans and plastic bottles located on a conveyor belt. An Oak-D RGB-D camera captures images of the scene, from which the 3D position and orientation of each item is calculated. With this information, the manipulator executes grasping actions to move the recyclable items to a recycling bin

After obtaining the segmented instances from the RGB image, Principal Component Analysis (PCA) (Greenacre et al., 2022) is performed on the mask of each item. PCA provides an efficient method to calculate different geometric properties from the object masks, and is first utilized to compute the centroid. The centroid represents the average position of all the pixels in the object mask, effectively providing a 2D estimate of the object’s center in the image plane. In addition to the centroid, PCA is utilized to calculate the principal axis of each mask, which is used to indicate the orientation of the each object in the image. Figure 5.2 provides a visual overview of this process, showing the initial RGB image, the extracted object masks, and the result of the PCA analysis on each mask.

The corresponding depth image is then used to recover the 3D position of the centroid from its 2D location in the image. Using the camera intrinsics and a standard pinhole model, the pixel corresponding to the centroid can be projected along a 3D ray. The depth value at the centroid’s pixel represents the

distance along this ray, allowing the computation of the object's centroid relative to the camera. This 3D position is transformed from the camera's local coordinate frame to the robot's global coordinate frame using ROS2 transformations.

The principal axis obtained from PCA defines the object's yaw angle in the robot frame. However, for a complete 6D pose representation, the pitch angle must also be estimated. To compute the pitch, a two-point method is employed. Two points are selected near opposite ends of the object along the principal axis. The 3D positions of these points are recovered from the depth image using the same strategy applied to the centroid. Let the vertical difference between the two points be Δz , and let the planar displacement in the xy -plane be Δxy . The pitch angle is then computed by $\omega = \arctan \frac{\Delta z}{\Delta xy}$, which provides a measure of the object's incline relative to the workspace plane. Due to assumed axial symmetry of the bottles and cans, the roll angle is assumed to be zero. Therefore, the full 6D pose of the each object is defined by the 3D position of the centroid and the estimated pitch and yaw angles.

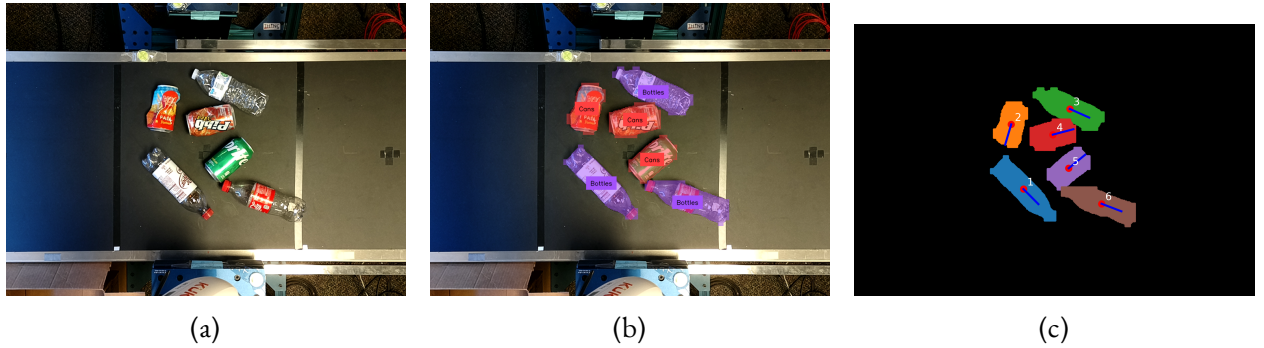


Figure 5.2: Context Estimation. Figure 5.2a shows the original RGB image. This image is passed through a YOLO v11 instance segmentation model to obtain masks of the bottles and cans, as shown in Figure 5.2b. Principal Component Analysis (PCA) is performed on the masks to determine the centroid and principal axes, shown in Figure 5.2c

5.1.3 Context Definition

Because the robot focuses on grasping one item at a time, the context representation only includes the 6D pose of the selected target. This approach greatly reduces the context space by concentrating on the object that is relevant for the current grasping action, rather than including the positions and orientations of all

other objects, which is difficult when the number of items is dynamic. Therefore, the context captures the position and orientation of the active object in 3D space, along with basic contextual information about the scene. Formally the context is defined as:

$$\mathbf{x} = [x, y, z, \omega, \psi, t, n],$$

where x , y , and z represent the Euclidian coordinates of the object’s centroid, ω and ψ denote the pitch and yaw angles describing the object’s orientation, t represents the object type (e.g., can or bottle), and n denotes the number of objects currently present in the scene. Information about the object type t is necessary as the geometry of aluminum cans and plastic bottles differs. Due to the physical constraints of the workspace, the following bounds are imposed on the context variables:

$$x \in [0.20, 0.55], y \in [-0.35, 0.35], z \in [0.80, 0.95],$$

$$\omega \in [-\frac{\pi}{2}, 0], \psi \in [-\pi, \pi].$$

These bounds ensure that the context representation includes only feasible poses, preventing the robot from considering impossible positions or orientations during planning or learning.

Because the true context space is continuous, there are infinitely many possible contexts. To significantly reduce the size of the context space for the reward and policy networks, a discretized encoding is introduced. The workspace is first divided into two equal partitions along the x and y axes, producing four spatial quadrants. Each object is assigned a quadrant based on the position of its centroid, compressing all possible continuous x and y values into four discrete values. Then, the pitch angle ω range is divided into three uniform bins, and the yaw angle ψ range is divided into four uniform bins. Like the quadrant representation, the pitch and yaw bins compress the context representation into a coarse but still meaningful form. The object type t is preserved in the discretized context to maintain the distinction between different recyclable items, while the number of objects n is removed to simplify the encoding

and because only a single target is considered. The resulting discretized state is compactly defined as

$$\hat{\mathbf{x}} = [q, \Omega, \Psi, t],$$

where $q \in \{1, 2, 3, 4\}$ denotes the spatial quadrant of the object, $\Omega \in \{1, 2, 3\}$ represents the discretized pitch bin, $\Psi \in \{1, 2, 3, 4\}$ denotes the discretized yaw bin, and t represents the object type. This encoding provides a simplified yet informative summary of the object and reduces the continuous context to a compact representation that is suitable for reward modeling and policy learning. Figure 5.3 shows the discretized workspace along with an example of an item mapped to the corresponding quadrant and orientation bins. Although an encoded, discretized context $\hat{\mathbf{x}}$ is produced, the true context $\mathbf{x}$ is preserved to build an absolute pose for the gripper from the relative action (see Section 5.1.4).

5.1.4 Action Definition

The actions for the sorting task define how the robotic manipulator moves prior to attempting a grasp. Rather than directly commanding joint angles or the absolute pose of the end effector in the world coordinate frame, each action specifies a relative displacement of the tool center point (TCP) or center of the gripper with respect to the pose of the target object. In other words, the action represents a position and orientation of the gripper relative to the currently selected item. In this formulation, the sole aim of the policy is to learn gripper configurations that lead to successful grasps, without reasoning about the dynamics or planning of the robotic arm. To fully capture the 6D pose of the gripper in 3D space, the action is defined as a 6-dimensional, continuous vector:

$$\mathbf{a} = [\Delta l, \Delta w, \Delta h, \Delta \varphi, \Delta \omega, \Delta \psi],$$

where $\Delta l, \Delta w, \Delta h$ denote relative Cartesian offsets in meters along the object's local x , y , and z axes, respectively, and $\Delta \varphi, \Delta \omega, \Delta \psi$ represent the relative rotations about the x , y , and z axes (roll, pitch, and yaw), respectively.

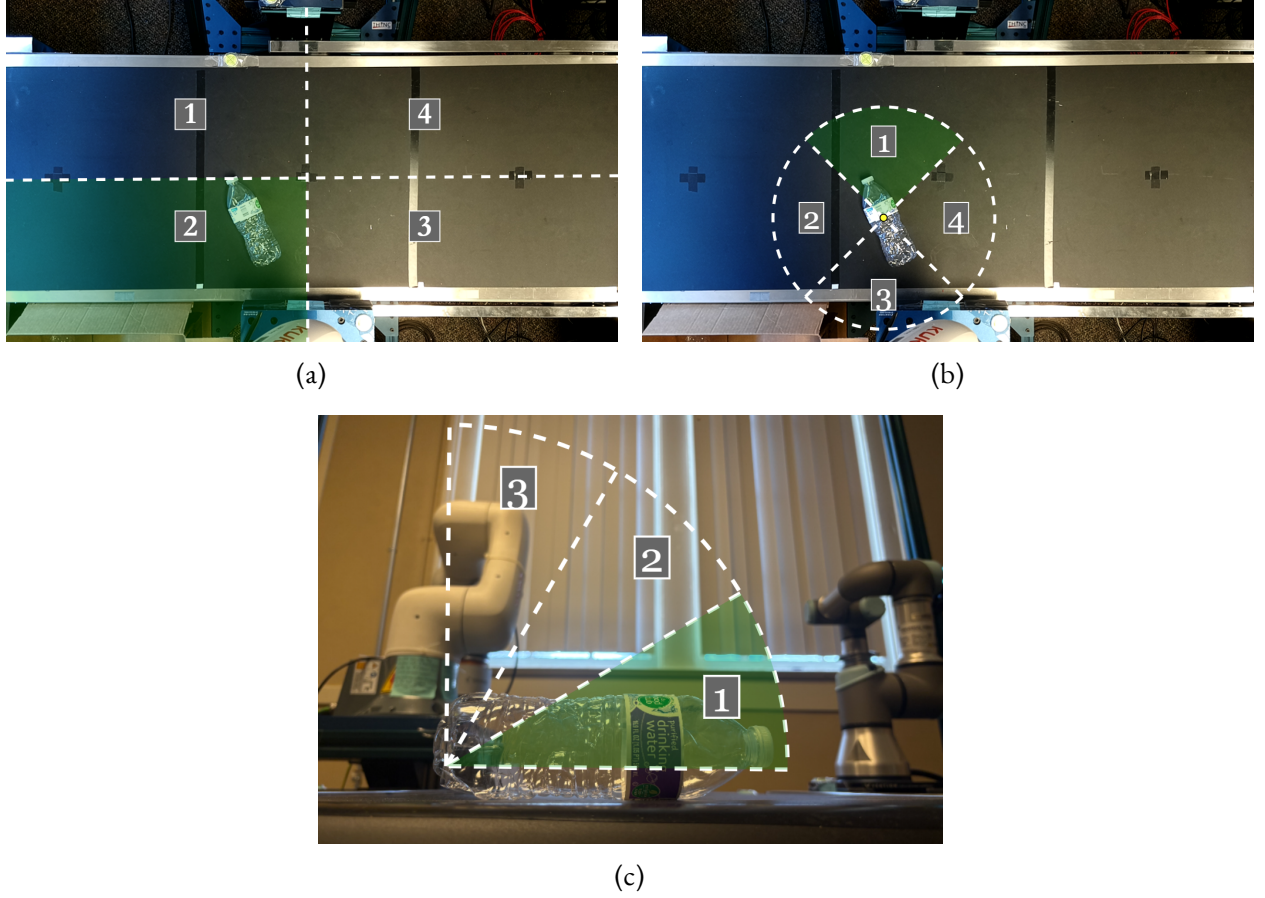


Figure 5.3: Discretized Workspace: Figure 5.3a shows the xy-plane of the workspace divided into four quadrants. In this example, the bottle is located in the second quadrant. Figure 5.3b illustrates the discretization of the possible yaw angles into four bins; the bottle falls within the first bin in this example. Similarly, the possible pitch angles are discretized into three bins, with the bottle located in the first bin, as shown in Figure 5.3c.

To execute an action, the agent first computes the absolute pose with component-wise addition between the relative action $\mathbf{a}$ and the preserved, true context $\mathbf{x}$. Once the absolute pose is determined, the agent uses an inverse kinematics (Calzada-Garcia et al., 2025) solver to compute a feasible joint configuration for the target pose. If a valid solution exists, a motion planner generates a trajectory that moves the manipulator from its starting state to the desired grasping position. If no valid solution is found, the action is considered infeasible and assigned a reward of zero.

For example, the action $[0.0, 0.0, 0.0, \frac{\pi}{4}, 0.0, 0.0]$ moves the TCP to the object’s centroid with matching pitch and yaw, while introducing a relative roll offset of $\frac{\pi}{4}$ radians or 45° . Figure 5.4 illustrates the object’s local axes and provides visual clarification of how each action component maps to translations and rotations about these axes. Because all components of the action are defined in the object’s local frame, the same relative action can produce different absolute motions depending on the object’s pose in the workspace. This introduces another element of stochasticity into the task, as the same action in the same context may lead to different results.

To ensure that the robot only attempts potentially successful grasps, each component of the relative action $\mathbf{a}$ has the following bounds

$$\Delta l \in [-0.10, 0.10], \Delta w \in [-0.10, 0.10], \Delta h \in [-0.01, 0.10],$$

$$\Delta \varphi \in [-\frac{\pi}{4}, \frac{\pi}{4}], \Delta \omega \in [-\frac{\pi}{4}, \frac{\pi}{4}], \Delta \psi \in [-\frac{\pi}{4}, \frac{\pi}{4}].$$

These bounds not only prevent the agent from proposing actions that will never lead to success, but they also shrink the action space significantly. In addition to these restrictions on the relative actions, the computed absolute pose is also clamped to the same bounds defined for the context (Section 5.1.3).

5.1.5 Reward Definition

Rewards are assigned autonomously to each action executed in the workspace. The reward is binary: actions that result in a successful grasp receive a reward of 1, while unsuccessful actions receive a reward of 0. Action success is determined using two criteria: a reduction in the total workspace items, and the measured current load of the gripper. If the number of items in the workspace decreases by one following the action and gripper current exceeds 150 mA, the action is classified as successful and assigned a reward of 1. Otherwise, the action is classified as a failure and assigned a reward of 0. This dual-condition check ensures that grasp attempts that fail to grab the object are not incorrectly labeled as successful.

A binary reward setup is effective for evaluating the method because multiple actions can yield successful grasps and achieve the maximum reward. However, many of these actions are suboptimal due to

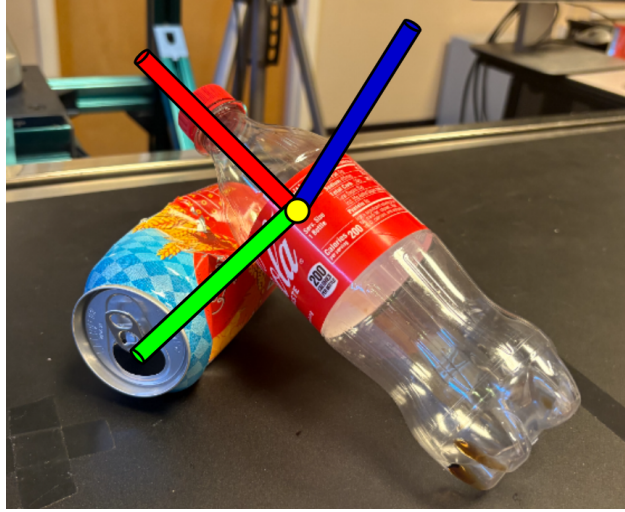


Figure 5.4: Local Coordinate Frame. The yellow dot on the bottle represents its centroid, while the red, green, and blue axes indicate the positive directions of the object’s local x , y , and z axes, respectively. An action containing a $+\Delta l$ component would produce a translation along the red axis, starting from the yellow dot.

inherent uncertainty in the workspace and the grasping process. Rather than optimizing for any action that simply achieves success, the modified objective encourages convergence towards actions that are more optimal and robust to uncertainty. These optimal actions are often located near the center of regions with consistently high rewards. Converging to these regions improves overall performance and helps avoid policies that settle in local optima or rely on highly situational actions.

5.2 Experimental Results

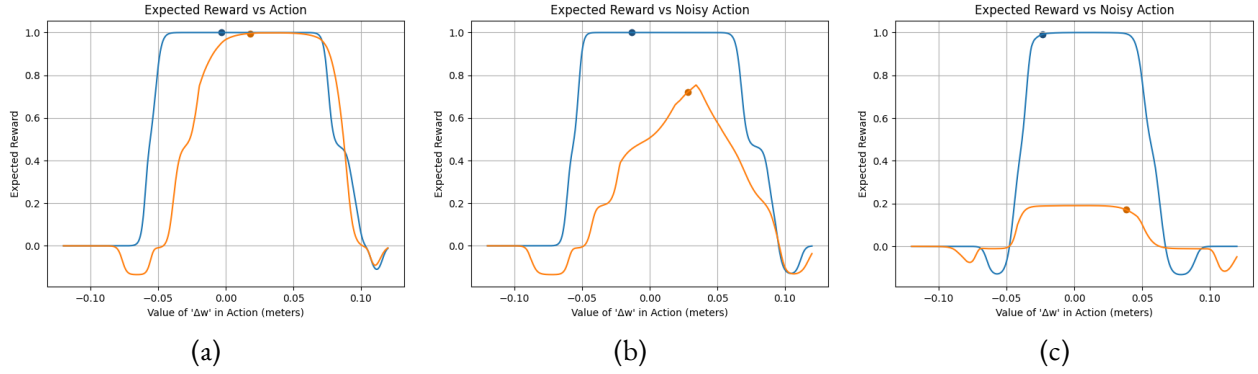


Figure 5.5: Expected Reward Comparisons: Each figure illustrates the expected reward $\mathcal{R}$ for a given state as the Δw component of the action is varied, comparing two policies trained on the same expected reward model. The blue lines correspond to the base action of the policy trained with the modified objective proposed in this work, while the orange lines show the same for the policy trained with the standard learning objective. The blue dot and orange dot indicate the respective base actions for each policy. Figure 5.5a directly compares the expected rewards of the base actions from both policies. Figures 5.5b and 5.5c show how the expected rewards change after applying the same adversarial perturbation to each action. Specifically, Figure 5.5b shows the effect of perturbing the action dimensions by 1 cm and 6° , while Figure 5.5c shows the effect of larger perturbations of 2 cm and 12° .

CHAPTER 6

CONCLUSION

6.1 Future Work

APPENDIX A

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